

Mutual Fund Performance Analytics Dashboard

Internship Project Report

Submitted By

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Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

Sonali Kumari

Certificate

(This page is reserved for the internship completion certificate. Please attach the certificate provided by Bluestock Fintech here.)

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Chapter 1: Introduction

1.1 Overview

Mutual funds are one of the most popular investment options because they allow investors to invest in professionally managed portfolios with relatively lower risk compared to investing directly in individual stocks. However, understanding the historical performance, risk level, and investment quality of a mutual fund requires detailed data analysis.

The Mutual Fund Performance Analytics Dashboard was developed to analyze historical Net Asset Value (NAV) data and convert raw financial information into meaningful business insights. The project combines Python-based data analysis with Power BI dashboard development to provide investors with an interactive platform for evaluating mutual fund performance.

The project begins with the collection of historical NAV data, followed by cleaning and preprocessing of the dataset. Various financial metrics such as Annualized Return, Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Rolling Sharpe Ratio, and Maximum Drawdown are then calculated to measure the fund's performance and risk characteristics.

To enhance the analysis, additional datasets such as investor profiles, sector allocation, fund metadata, and recommendation data were generated for educational and analytical purposes. These datasets enabled advanced analytics, including investor segmentation, sector analysis, SIP analysis, and fund recommendation.

Finally, an interactive Power BI dashboard was created to visualize the analytical results using KPI cards, charts, filters, and business insights. The dashboard allows users to explore financial performance efficiently and supports better investment decision-making.

1.2 Importance of the Project

The increasing popularity of mutual fund investments has created a need for analytical tools that simplify financial decision-making. Investors often struggle to interpret raw NAV data and compare different investment opportunities.

This project addresses this challenge by providing an integrated analytics solution that transforms historical financial data into easy-to-understand visual reports. It enables users to evaluate fund performance, identify market trends, assess investment risk, and explore sector-wise and investor-wise insights through interactive dashboards.

The project also demonstrates how data analytics, business intelligence, and financial analysis can be combined to solve real-world investment problems.

1.3 Scope of the Project

The scope of this project includes:

- Collection and preprocessing of historical mutual fund NAV data.
- Cleaning and transformation of financial datasets using Python.
- Calculation of important performance indicators such as CAGR, Annualized Return, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.
- Generation of supporting datasets for investor analysis, fund metadata, sector allocation, and recommendation systems.
- Development of interactive Power BI dashboards for business users.
- Presentation of meaningful insights to support investment analysis.

Although the project focuses on historical data analysis, it establishes a strong foundation for future enhancements such as live market integration, predictive analytics, and machine learning-based fund recommendations.

Chapter 2: Project Objectives

2.1 Primary Objective

The primary objective of this project is to develop a comprehensive Mutual Fund Performance Analytics Dashboard that transforms raw historical Net Asset Value (NAV) data into meaningful financial insights using Python, SQL, and Power BI. The dashboard aims to help investors understand fund performance, evaluate investment risk, and make informed financial decisions through interactive visualizations.

2.2 Specific Objectives

The major objectives of this project are as follows:

- To collect and preprocess historical mutual fund NAV data.
- To clean and transform the dataset for accurate analysis.
- To calculate important financial performance metrics such as Annualized Return, Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Maximum Drawdown, and Rolling Sharpe Ratio.
- To perform Exploratory Data Analysis (EDA) for identifying historical trends and investment patterns.
- To generate additional datasets, including investor information, sector allocation, fund metadata, and recommendation data, for advanced business analysis.
- To build an interactive Power BI dashboard containing KPI cards, trend charts, slicers, and business insights.
- To provide investors with a user-friendly platform for analyzing mutual fund performance and comparing investment opportunities.

2.3 Expected Outcomes

The expected outcomes of this project include:

- A cleaned and structured financial dataset ready for analysis.
- Automated calculation of performance and risk metrics.
- Interactive Power BI dashboards for visual analysis.
- Business insights to support investment decision-making.
- A scalable analytics framework that can be extended with live financial data in the future.

Chapter 3: Problem Statement

3.1 Background

Mutual fund investors often rely on raw historical NAV data, which is difficult to interpret without proper analysis. Although financial websites provide large amounts of investment information, they rarely present comprehensive analytical insights in an interactive and easy-to-understand format.

As a result, investors may find it challenging to compare mutual funds, evaluate risk, monitor historical performance, or identify suitable investment options based on their financial goals.

3.2 Problem Definition

The project addresses the challenge of converting historical mutual fund NAV data into meaningful business intelligence. The goal is to build an analytics system capable of calculating financial performance indicators, generating business insights, and presenting the results through interactive dashboards.

The solution simplifies investment analysis by integrating data preprocessing, financial metric calculation, advanced analytics, and visualization into a single platform.

3.3 Proposed Solution

To overcome these challenges, the project implements an end-to-end mutual fund analytics pipeline consisting of:

- Historical NAV data preprocessing using Python.
- Financial performance calculations.
- Generation of analytical datasets for investors, sectors, and recommendations.
- SQL-based data management.
- Interactive dashboards using Power BI.

- Business insights for better investment decision-making.

This integrated approach enables users to evaluate mutual fund performance efficiently while reducing the complexity of manual financial analysis.

Chapter 4: Dataset Description

4.1 Original Dataset

The primary dataset used in this project consists of historical Net Asset Value (NAV) records of a mutual fund scheme. The dataset contains daily observations representing the market value of the fund over multiple years.

Original Attributes

Column Name	Description
Date	Trading date of the NAV record
NAV	Net Asset Value of the mutual fund

4.2 Generated Supporting Datasets

To perform advanced business analytics, several supporting datasets were generated using realistic assumptions and financial logic.

1. Performance Summary

Contains calculated financial performance indicators such as Annualized Return, CAGR, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.

2. Rolling Metrics

Stores rolling return, rolling volatility, and rolling Sharpe Ratio values used for time-series analysis.

3. Drawdown Data

Contains historical drawdown values representing declines from previous NAV peaks.

4. Investor Dataset

Includes simulated investor information such as age group, SIP amount, investment category, and risk profile.

5. Fund Metadata

Contains additional information about mutual funds, including category, expense ratio, expected return, risk level, and Assets Under Management (AUM).

6. Sector Allocation

Represents the allocation of investments across different industrial sectors.

7. Recommended Funds

Provides sample fund recommendations based on category, expected return, risk profile, and expense ratio.

4.3 Purpose of Dataset Generation

The original NAV dataset alone was insufficient for advanced analytics. Therefore, additional datasets were generated to support:

- Investor segmentation
- Fund recommendation
- Sector-wise analysis
- Risk assessment
- Interactive dashboard creation
- Business intelligence reporting

These datasets simulate realistic financial scenarios while maintaining consistency with the original NAV data.

Chapter 5: Technology Stack

5.1 Overview

This project was developed using a combination of programming languages, libraries, and business intelligence tools suited for financial data analytics. The workflow covers data ingestion, preprocessing, database management, financial performance calculations, advanced risk analytics, and interactive dashboard development.

5.2 Technologies Used

- Python — used for data ingestion, cleaning, preprocessing, and financial metric calculations.
- Pandas and NumPy — used for data manipulation, transformation, and numerical computation.
- Matplotlib — used for visualizing trends during Exploratory Data Analysis.
- SQL and SQLite — used for structured data storage and querying (database file: `bluestock_mf.db`).
- Jupyter Notebook — used for exploratory analysis and performance/advanced analytics workflows.
- Visual Studio Code — used as the primary code editor for scripts and notebooks.
- Microsoft Power BI — used to build the interactive analytics dashboard.
- Git — used for version control of the project codebase.

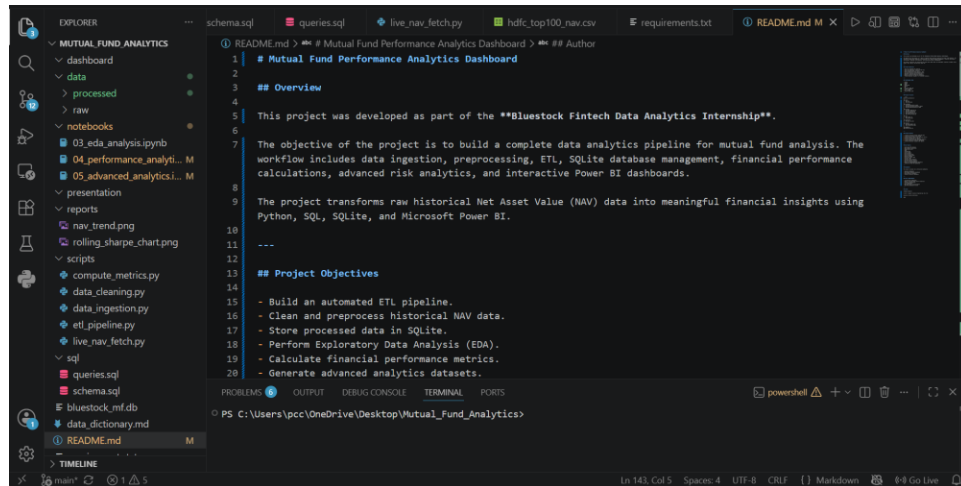


Figure 5.1: Project README Describing Objectives and Technology Stack

5.3 Project Structure

The project follows an organized folder structure that separates raw data, processed data, notebooks, scripts, SQL files, and dashboard files. This structure keeps the data pipeline, analysis, and reporting components clearly separated and easy to maintain.

- dashboard/ — contains the Power BI dashboard file.
- data/raw/ and data/processed/ — store the original and cleaned NAV datasets.
- notebooks/ — contains the EDA, performance analytics, and advanced analytics notebooks.
- scripts/ — contains Python scripts for data ingestion, cleaning, the ETL pipeline, live NAV fetching, and metric computation.
- sql/ — contains schema and query files for the SQLite database.
- reports/ — stores exported chart images generated during analysis.
- README.md and data_dictionary.md — document the project overview and dataset fields.

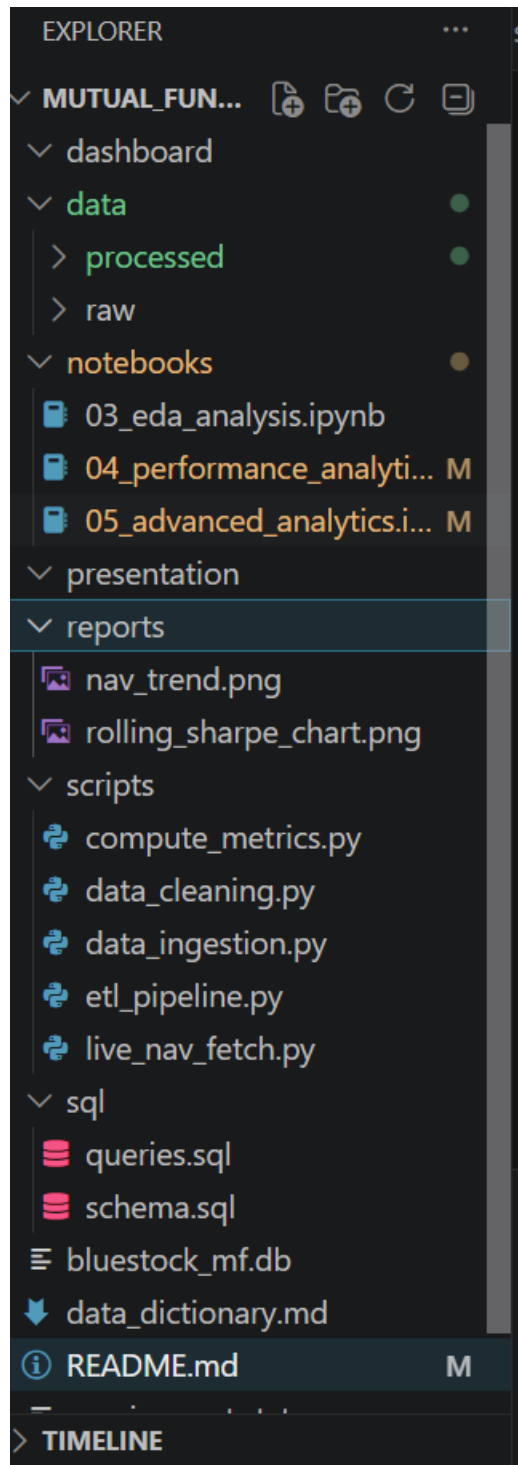


Figure 5.2: Project Folder Structure in Visual Studio Code

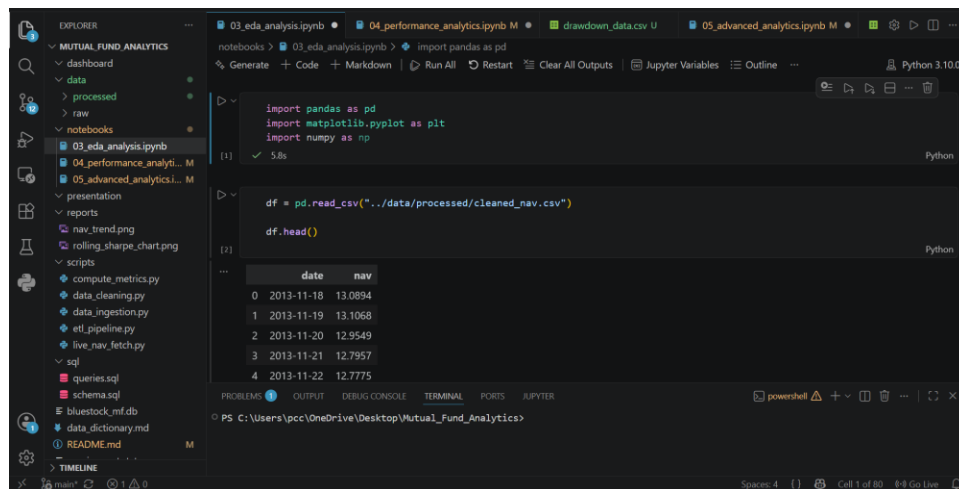
Chapter 6: Data Preprocessing

6.1 Overview

Before any financial metrics could be calculated, the raw historical NAV data was ingested and cleaned using Python. This chapter describes the preprocessing pipeline that converts raw data into an analysis-ready format.

6.2 Data Ingestion and Cleaning

Raw NAV data was ingested using the `data_ingestion.py` script and cleaned using `data_cleaning.py`. The cleaning process handled missing or invalid values and formatted the date field consistently, producing a structured dataset ready for analysis.



The screenshot shows a Jupyter Notebook environment with a file explorer on the left and a code editor on the right. The file explorer displays a project structure for 'MUTUAL_FUND_ANALYTICS' with folders for 'dashboard', 'data', 'processed', 'raw', 'notebooks', 'presentation', 'reports', 'nav_trend.png', 'rolling_shape_chart.png', 'scripts', 'sql', 'queries.sql', 'schema.sql', 'bluestock_mf.db', 'data_dictionary.md', and 'README.md'. The 'notebooks' folder is expanded, showing '03_eda_analysis.ipynb' as the active notebook. The code editor shows two cells of Python code. Cell [1] imports pandas, matplotlib, and numpy. Cell [2] reads a CSV file and displays the first five rows of the resulting DataFrame.

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```
df = pd.read_csv("../data/processed/cleaned_nav.csv")
df.head()
```

	date	nav
0	2013-11-18	13.0894
1	2013-11-19	13.1068
2	2013-11-20	12.9549
3	2013-11-21	12.7957
4	2013-11-22	12.7775

Figure 6.1: Loading the Cleaned NAV Dataset in the EDA Notebook

6.3 Daily Return Calculation and Export

The daily percentage change in NAV was calculated using pandas' `pct_change()` function to derive the `daily_return` column. The resulting dataset, including date, NAV, and daily return values, was exported to a processed CSV file (`cleaned_nav.csv`) for use in subsequent performance and risk calculations.

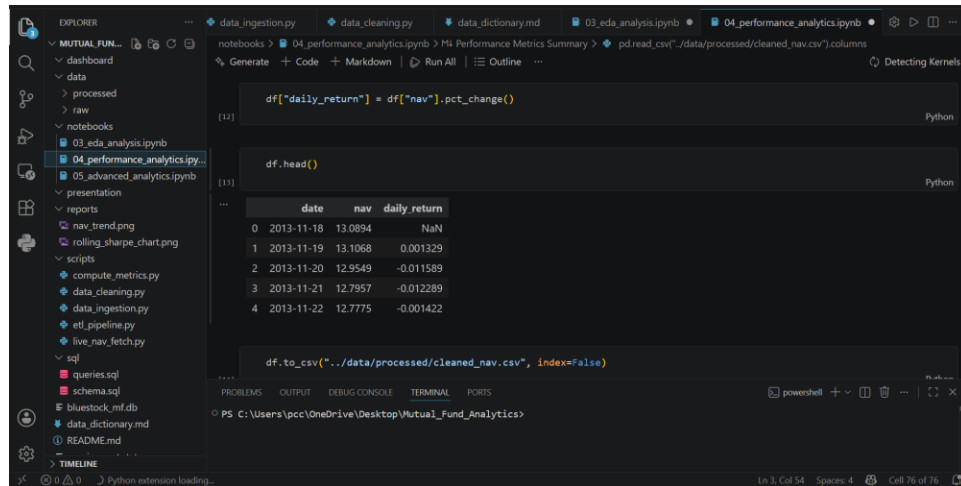


Figure 6.3: Daily Return Calculation and Processed Data Export

6.4 Database Integration

The cleaned dataset was integrated with a SQLite database (`bluestock_mf.db`) using schema and query definitions stored in `schema.sql` and `queries.sql`. This enabled structured storage and efficient retrieval of NAV records for downstream analytics.

Chapter 7: Exploratory Data Analysis (EDA)

7.1 Overview

Exploratory Data Analysis (EDA) was performed to understand the historical behavior of the mutual fund before applying financial analytics. The objective was to identify trends, detect anomalies, understand data distribution, and validate the dataset for further analysis.

Python libraries such as Pandas and Matplotlib were used to visualize the historical NAV data and generate meaningful statistical insights.

7.2 Historical NAV Trend

The historical NAV trend demonstrates the long-term performance of the mutual fund. The analysis shows that despite short-term market fluctuations, the overall NAV exhibits a consistent upward trend over the investment period, indicating stable long-term growth.

7.3 Daily Return Analysis

Daily returns were calculated using the percentage change in NAV values. This analysis helps identify periods of high market volatility and provides the basis for calculating advanced financial metrics such as Sharpe Ratio and Rolling Return.

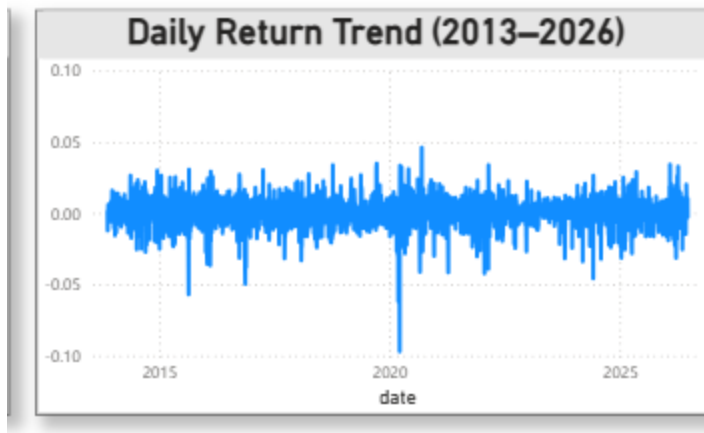


Figure 7.1: Daily Return Visualization

7.4 Statistical Summary

The dataset was analyzed using descriptive statistics, including mean, median, minimum, maximum, and standard deviation. These statistics provide an overall understanding of the central tendency

7.5 Observations

and variability of the mutual fund's performance.

- Historical NAV shows long-term positive growth.
- Daily returns fluctuate based on market conditions.
- The dataset contains sufficient observations for performance analysis.
- Cleaned data was suitable for advanced financial calculations.

Chapter 8: Performance Analytics

8.1 Overview

Performance Analytics evaluates the historical returns and risk-adjusted performance of the mutual fund using standard financial metrics.

The calculations were performed using Python and later visualized in Power BI.

8.2 Financial Metrics Calculated

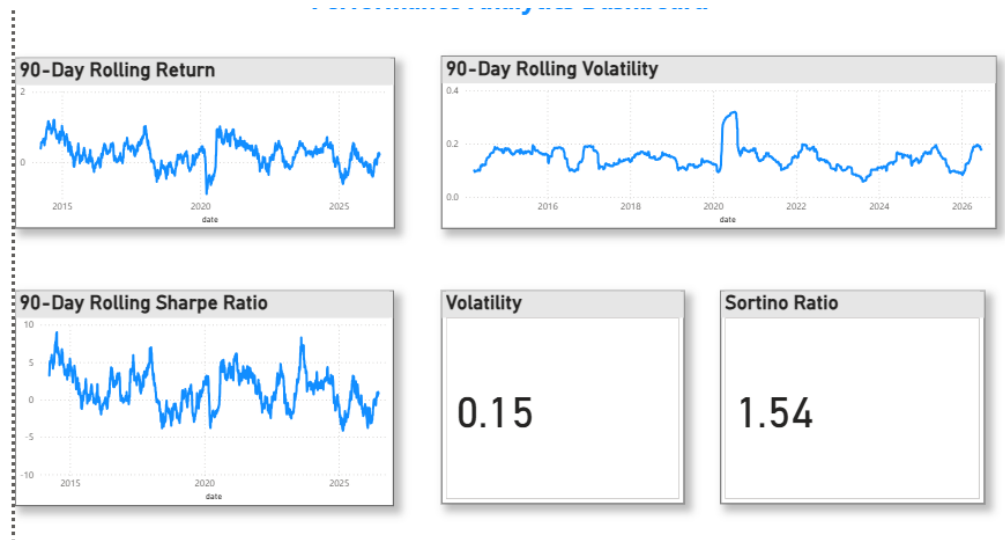
- Annualized Return
- Compound Annual Growth Rate (CAGR)
- Volatility
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Rolling Return
- Rolling Volatility
- Rolling Sharpe Ratio

8.3 Rolling Performance Analysis

Rolling metrics provide insights into changing fund performance over time. The project calculated:

- 90-Day Rolling Return
- 90-Day Rolling Volatility
- 90-Day Rolling Sharpe Ratio

These indicators help investors monitor consistency and risk-adjusted returns throughout different market periods.



8.4 Risk Indicators

Risk metrics such as Maximum Drawdown and Volatility help investors understand downside risk and market uncertainty.

These indicators improve investment decision-making by highlighting periods of increased financial risk.

Chapter 9: Advanced Analytics

9.1 Overview

To extend the analytical capabilities of the project, multiple supporting datasets were generated using realistic financial assumptions. These datasets enabled investor analysis, sector analysis, recommendation systems, and advanced business intelligence.

9.2 Investor Analytics

The investor dataset was created to simulate different investment profiles based on age, investment amount, SIP category, and risk preference.

This dataset supports investor segmentation and behavioral analysis.

9.3 Sector Allocation

Sector allocation analysis visualizes the distribution of investments across various industry sectors. Diversification reduces concentration risk and provides a balanced investment portfolio.

9.4 Mutual Fund Recommendation

A recommendation dataset was generated to suggest suitable mutual funds based on investment category, expected return, expense ratio, and investor risk profile.

9.5 Advanced Risk Metrics

Additional financial indicators were calculated to evaluate portfolio risk. These include:

- Historical Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Portfolio Herfindahl-Hirschman Index (HHI)

These metrics provide deeper insights into portfolio diversification and downside risk.

Chapter 10: Dashboard Implementation

10.1 Overview

Microsoft Power BI was used to develop an interactive dashboard for visualizing financial performance, investment risk, and investor insights. The dashboard integrates multiple processed datasets and provides an intuitive interface for exploring analytical results.

10.2 Executive Dashboard

The Executive Dashboard presents the overall performance of the mutual fund through KPI cards and trend visualizations.

Features include:

- Annualized Return
- CAGR
- Sharpe Ratio
- Maximum Drawdown
- Historical NAV Trend
- Daily Return Trend

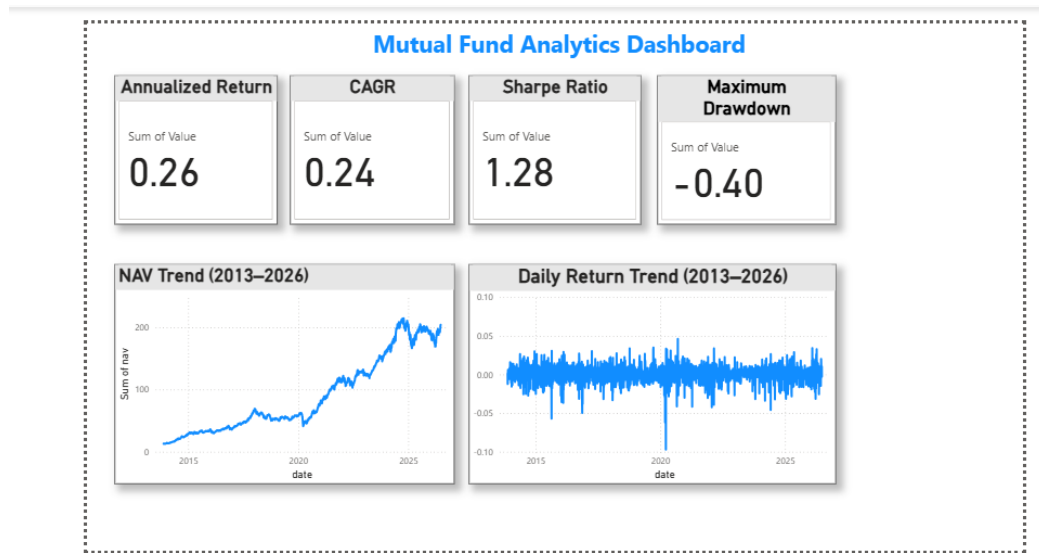


Figure 10.1: Executive Dashboard

10.3 Performance Analytics Dashboard

This dashboard focuses on rolling performance indicators, including:

- Rolling Return
- Rolling Volatility
- Rolling Sharpe Ratio
- Volatility
- Sortino Ratio

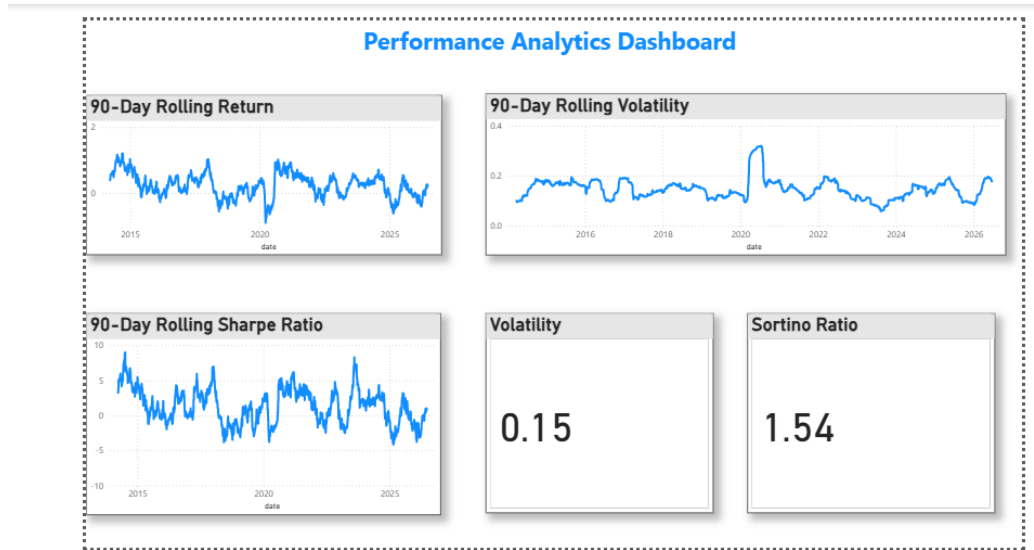


Figure 10.2: Performance Analytics Dashboard

10.4 Risk Analytics Dashboard

The Risk Dashboard provides insights into investment risk through:

- Historical VaR
- Historical CVaR
- Portfolio HHI
- Maximum Drawdown Trend
- Sector Allocation

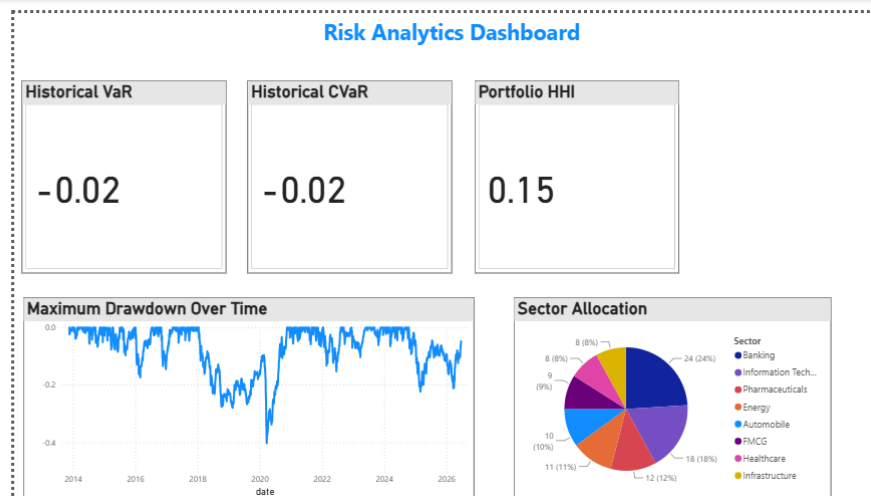


Figure 10.3: Risk Analytics Dashboard

10.5 Investor Analytics Dashboard

The Investor Dashboard includes:

- Investor Cohort Analysis
- SIP Investment Distribution
- Recommended Mutual Funds
- Risk Level Filtering

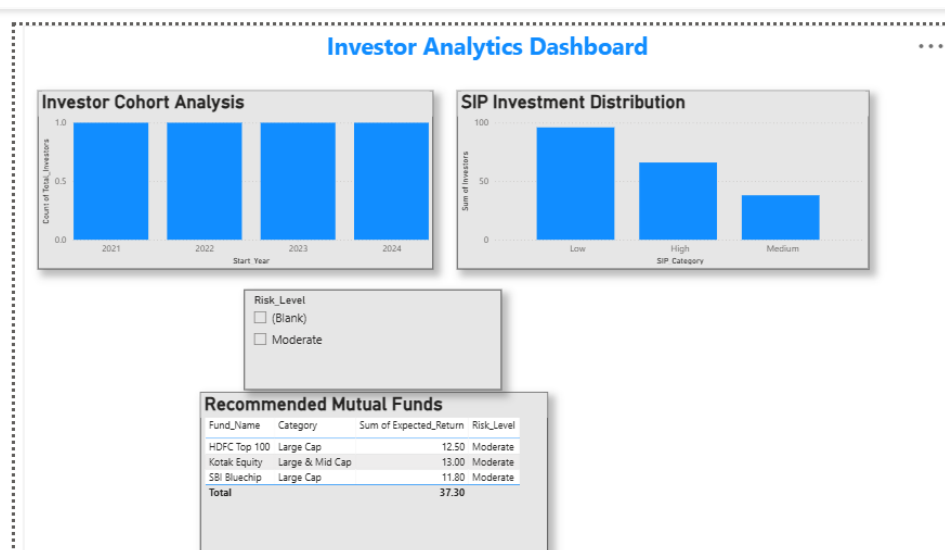


Figure 10.4: Investor Analytics Dashboard

10.6 Dashboard Benefits

The developed dashboard enables users to:

- Monitor mutual fund performance.
- Evaluate investment risk.
- Analyze historical trends.
- Explore investor behavior.
- Support informed investment decisions.

Chapter 11: Business Insights

11.1 Overview

The developed analytics system successfully transformed historical financial data into meaningful business insights. The Power BI dashboard enables users to understand mutual fund performance, evaluate investment risk, and identify investment opportunities through interactive visualizations.

11.2 Key Business Insights

Long-Term Growth

Historical NAV analysis indicates that the mutual fund demonstrates consistent long-term appreciation despite periodic market fluctuations.

Risk-Adjusted Returns

The Sharpe Ratio and Sortino Ratio indicate that the mutual fund delivers reasonable returns relative to investment risk.

Market Volatility

Rolling Volatility analysis highlights periods of increased market uncertainty, helping investors identify changing market conditions.

Downside Risk

Maximum Drawdown, Historical VaR, and CVaR provide investors with valuable information regarding potential financial losses during adverse market scenarios.

Portfolio Diversification

Sector Allocation analysis demonstrates that investments are distributed across multiple industries, reducing concentration risk.

Investor Segmentation

Investor Cohort Analysis and SIP Distribution provide useful insights into investor behavior and investment preferences.

Fund Recommendation

The recommendation dashboard simplifies investment selection by allowing investors to compare funds based on category, expected return, and risk profile.

Chapter 12: Conclusion

12.1 Conclusion

The **Mutual Fund Performance Analytics Dashboard** was successfully developed to analyze historical mutual fund performance using Python, SQLite, SQL, and Microsoft Power BI. The project transformed raw NAV data into meaningful financial insights through data preprocessing, ETL, financial metric calculations, and interactive dashboards.

The implementation included performance analysis, risk assessment, investor analytics, and fund recommendation. Four Power BI dashboards were created to visualize KPIs, rolling metrics, risk indicators, and investor information in an easy-to-understand format.

Overall, the project demonstrates how data analytics and business intelligence techniques can support informed investment decisions and improve the understanding of mutual fund performance.

12.2 Project Outcomes

The project successfully achieved the following outcomes:

- Developed an automated ETL pipeline.
- Generated cleaned and structured datasets.

- Calculated key financial performance metrics.
- Performed advanced financial analytics.
- Designed four interactive Power BI dashboards.
- Generated business insights for investment analysis.

Chapter 13: Future Scope

The project can be further enhanced with additional features to improve functionality and usability.

Possible future improvements include:

- Integration with live Mutual Fund APIs for real-time NAV updates.
- Machine Learning models for return prediction.
- AI-based personalized investment recommendations.
- Portfolio optimization techniques.
- Cloud deployment using Azure or AWS.
- Streamlit or Flask web application.
- Automated PDF report generation.
- Mobile-friendly dashboard interface.
- Benchmark comparison with NIFTY and Sensex indices.
- Email-based investment alerts and notifications.

These enhancements would make the system more intelligent, scalable, and suitable for real-world financial applications.

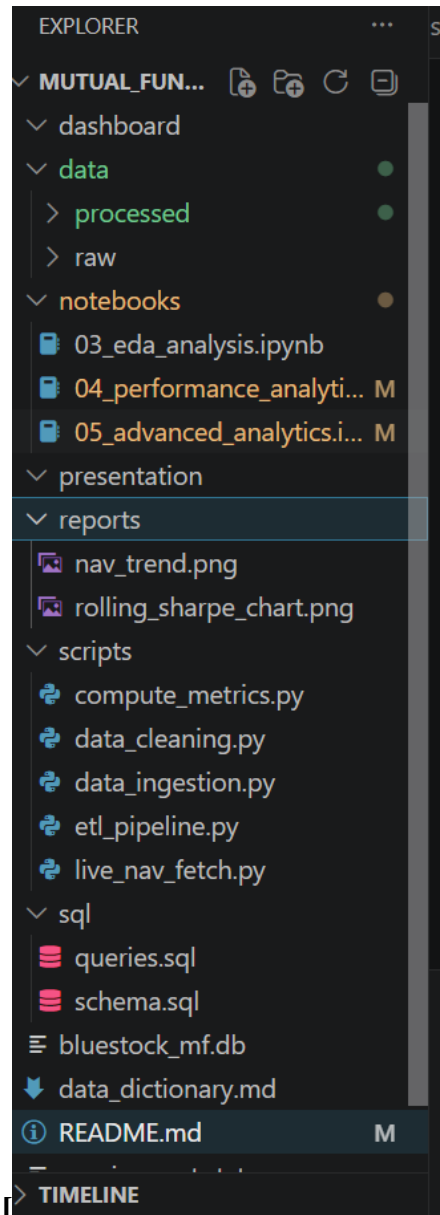
REFERENCES

The following resources were used during the development of this project:

1. Microsoft Power BI Documentation.
2. Python Official Documentation.
3. Pandas Documentation.
4. NumPy Documentation.
5. Matplotlib Documentation.
6. SQLite Documentation.
7. Microsoft Learn – Power BI.
8. Investopedia – Mutual Fund Concepts.
9. SEBI (Securities and Exchange Board of India).
10. National Stock Exchange (NSE) India.

APPENDIX

Appendix A – Project Folder Structure



Appendix B – Jupyter Notebook Outputs

Insert screenshots of:

- ETL Notebook
- EDA Notebook
- Performance Analytics Notebook
- Advanced Analytics Notebook
- Dataset Generation Notebook

Appendix C – Power BI Dashboards

Insert screenshots of:

- Executive Dashboard
- Performance Analytics Dashboard
- Risk Analytics Dashboard
- Investor Analytics Dashboard